Practical 1. Predicting House Price using Multiple Linear Regression

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score

# Step 1: Load the dataset
data = pd.read_csv("house_price.csv")
print("First 5 rows of dataset:")
print(data.head())

# Step 2: Define independent (X) and dependent (y) variables
X = data[['Area_sqft', 'Bedrooms', 'Age_years']]
y = data['Price_in_thousands']

# Step 3: Split data into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42)

# Step 4: Create and train the Multiple Linear Regression model
model = LinearRegression()
model.fit(X_train, y_train)

# Step 5: Predict on test data
y_pred = model.predict(X_test)

# Step 6: Display model parameters
print("\nIntercept (b0):", model.intercept_)
print("Coefficients (b1, b2, b3):", model.coef_)

# Step 7: Evaluate the model
print("\nActual Values:", list(y_test))
print("Predicted Values:", list(y_pred))
print("\nMean Squared Error:", mean_squared_error(y_test, y_pred))
print("R2 Score:", r2_score(y_test, y_pred))

# Step 8: Predict price for a new house
new_house = pd.DataFrame([[1800, 3, 10]], columns=['Area_sqft', 'Bedrooms', 'Age_years'])
predicted_price = model.predict(new_house)
print("\nPredicted Price for new house (in thousands):", predicted_price[0])
```

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Output:

First 5 rows of dataset:

	Area_sqft	Bedrooms	Age_years	Price_in_thousands
0	1661	2	12	433.66
1	835	2	26	390.27
2	1696	2	20	390.22
3	1505	1	16	207.41
4	1560	5	13	1004.64

Intercept (b0): -31.241609288412178
Coefficients (b1, b2, b3): [7.29286813e-02 1.96658301e+02 -3.02946236e+00]

Actual Values: [433.66, 178.65, 969.97, 390.27]
Predicted Values: [446.8559838301919, 214.38817757093378, 889.3046629695498, 344.2044199811176]

Mean Squared Error: 2520.0713965126156
R2 Score: 0.9703861498527574

Predicted Price for new house (in thousands): 659.7102961753193

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Practical 2. Predicting Student Exam Score using Multiple Linear Regression

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score

# Step 1: Load the dataset
data = pd.read_csv("student_score.csv")
print("First 5 rows of dataset:")
print(data.head())

# Step 2: Define independent (X) and dependent (y) variables
X = data[['Hours_Studied', 'Attendance_Percent', 'Previous_Score']]
y = data['Exam_Score']

# Step 3: Split data into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42)

# Step 4: Create and train the Multiple Linear Regression model
model = LinearRegression()
model.fit(X_train, y_train)

# Step 5: Predict on test data
y_pred = model.predict(X_test)

# Step 6: Display model parameters
print("\nIntercept (b0):", model.intercept_)
print("Coefficients (b1, b2, b3):", model.coef_)

# Step 7: Evaluate the model
print("\nActual Values:", list(y_test))
print("Predicted Values:", list(y_pred))
print("\nMean Squared Error:", mean_squared_error(y_test, y_pred))
print("R2 Score:", r2_score(y_test, y_pred))

# Step 8: Predict exam score for a new student
new_student = pd.DataFrame([[6, 85, 70]],
    columns=['Hours_Studied', 'Attendance_Percent', 'Previous_Score'])
predicted_score = model.predict(new_student)
print("\nPredicted Exam Score for new student:", predicted_score[0])
```

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Result / Output:

First 5 rows of dataset:

	Hours_Studied	Attendance_Percent	Previous_Score	Exam_Score
0	4.9	54	85	72.2
1	1.2	96	48	63.3
2	5.9	56	57	69.2
3	4.9	81	62	77.3
4	4.8	69	93	78.8

Intercept (b0): -10.500929526815199
Coefficients (b1, b2, b3): [5.48009451 0.4757004 0.33489934]

Actual Values: [72.2, 77.4, 84.5, 63.3]
Predicted Values: [70.5057984942311, 81.02808521820569, 95.63024212849994, 57.81758998581225]

Mean Squared Error: 42.49310762385332
R2 Score: 0.2889670341124734

Predicted Exam Score for new student: 86.25712468601384

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Practical 3. Predicting Employee Salary using Multiple Linear Regression

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score

# Step 1: Load the dataset
data = pd.read_csv("employee_salary.csv")
print("First 5 rows of dataset:")
print(data.head())

# Step 2: Define independent (X) and dependent (y) variables
X = data[['Experience_Years', 'Education_Level', 'Age']]
y = data['Salary']

# Step 3: Split data into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42)

# Step 4: Create and train the Multiple Linear Regression model
model = LinearRegression()
model.fit(X_train, y_train)

# Step 5: Predict on test data
y_pred = model.predict(X_test)

# Step 6: Display model parameters
print("\nIntercept (b0):", model.intercept_)
print("Coefficients (b1, b2, b3):", model.coef_)

# Step 7: Evaluate the model
print("\nActual Values:", list(y_test))
print("Predicted Values:", list(y_pred))
print("\nMean Squared Error:", mean_squared_error(y_test, y_pred))
print("R2 Score:", r2_score(y_test, y_pred))

# Step 8: Predict salary for a new employee
new_employee = pd.DataFrame([[7, 2, 30]],
    columns=['Experience_Years', 'Education_Level', 'Age'])
predicted_salary = model.predict(new_employee)
print("\nPredicted Salary for new employee:", predicted_salary[0])
```

Result / Output:

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First 5 rows of dataset:

	Experience_Years	Education_Level	Age	Salary
0	11.0	2	29	56675.79
1	14.2	3	23	82104.88
2	5.8	1	43	31436.49
3	10.2	3	32	68383.16
4	17.9	1	49	78357.34

Intercept (b0): 9931.250875945145
Coefficients (b1, b2, b3): [3272.92045812 9475.80496623 -42.75145629]

Actual Values: [56675.79, 57696.06, 70782.1, 82104.88]
Predicted Values: [63645.193615189084, 54011.69507100557, 75953.09222938324, 83850.85278515547]

Mean Squared Error: 22983678.32106503
R2 Score: 0.7888873074398451

Predicted Salary for new employee: 50510.76032643086

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Practical 4.Predicting Used Car Price using Multiple Linear Regression

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score

# Step 1: Load the dataset
data = pd.read_csv("car_price.csv")
print("First 5 rows of dataset:")
print(data.head())

# Step 2: Define independent (X) and dependent (y) variables
X = data[['Mileage_km', 'Age_years', 'Engine_Size_L']]
y = data['Price']

# Step 3: Split data into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42)

# Step 4: Create and train the Multiple Linear Regression model
model = LinearRegression()
model.fit(X_train, y_train)

# Step 5: Predict on test data
y_pred = model.predict(X_test)

# Step 6: Display model parameters
print("\nIntercept (b0):", model.intercept_)
print("Coefficients (b1, b2, b3):", model.coef_)

# Step 7: Evaluate the model
print("\nActual Values:", list(y_test))
print("Predicted Values:", list(y_pred))
print("\nMean Squared Error:", mean_squared_error(y_test, y_pred))
print("R2 Score:", r2_score(y_test, y_pred))

# Step 8: Predict price for a new car
new_car = pd.DataFrame([[45000, 4, 2.0]],
    columns=['Mileage_km', 'Age_years', 'Engine_Size_L'])
predicted_price = model.predict(new_car)
print("\nPredicted Price for new car:", predicted_price[0])
```

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Result / Output:

First 5 rows of dataset:

	Mileage_km	Age_years	Engine_Size_L	Price
0	88066	3	3.2	946972.18
1	78902	0	1.4	750512.24
2	97599	7	1.2	586902.03
3	16863	5	2.5	1024949.71
4	28346	5	1.4	832138.13

Intercept (b0): 808148.3014711478
Coefficients (b1, b2, b3): [-2.84082903e+00 -2.15179785e+04 1.47373454e+05]

Actual Values: [946972.18, 573154.57, 739154.06, 750512.24]
Predicted Values: [965008.9712377572, 596320.2290632488, 761948.7343776461, 790324.0457589075]

Mean Squared Error: 741637663.9392529
R2 Score: 0.9577211272986046

Predicted Price for new car: 888985.9903230164

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Practical 5. Predicting Crop Yield using Multiple Linear Regression

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score

# Step 1: Load the dataset
data = pd.read_csv("crop_yield.csv")
print("First 5 rows of dataset:")
print(data.head())

# Step 2: Define independent (X) and dependent (y) variables
X = data[['Rainfall_mm', 'Fertilizer_kg_per_acre', 'Temperature_C']]
y = data['Crop_Yield_quintal_per_acre']

# Step 3: Split data into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42)

# Step 4: Create and train the Multiple Linear Regression model
model = LinearRegression()
model.fit(X_train, y_train)

# Step 5: Predict on test data
y_pred = model.predict(X_test)

# Step 6: Display model parameters
print("\nIntercept (b0):", model.intercept_)
print("Coefficients (b1, b2, b3):", model.coef_)

# Step 7: Evaluate the model
print("\nActual Values:", list(y_test))
print("Predicted Values:", list(y_pred))
print("\nMean Squared Error:", mean_squared_error(y_test, y_pred))
print("R2 Score:", r2_score(y_test, y_pred))

# Step 8: Predict crop yield for new field conditions
new_field = pd.DataFrame([[200, 60, 28]],
    columns=['Rainfall_mm', 'Fertilizer_kg_per_acre', 'Temperature_C'])
predicted_yield = model.predict(new_field)
print("\nPredicted Crop Yield (quintal/acre):", predicted_yield[0])
```

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Result / Output:

First 5 rows of dataset:

	Rainfall_mm	...	Crop_Yield_quintal_per_acre
0	149	...	59.55
1	256	...	94.85
2	289	...	59.85
3	239	...	63.87
4	280	...	94.32

[5 rows x 4 columns]

Intercept (b0): 30.689807420289988
Coefficients (b1, b2, b3): [0.14603824 0.57219144 -0.95774278]

Actual Values: [59.55, 53.23, 39.25, 94.85]
Predicted Values: [55.45543974889065, 51.62963097542454, 39.70416686876836, 86.33915147265674]

Mean Squared Error: 22.99185371621325
R2 Score: 0.9452342417835637

Predicted Crop Yield (quintal/acre): 67.41214353286381

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